

WASP-11b

R-0867

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_181207 · created 2026-07-28T22:52:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458459.8285 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1 +/- 0.049
Transit depth (Rp/Rs)^2	1.0 +/- 0.99 [%]
Orbital Inclination (inc)	86.35 +/- 1.7
Airmass coefficient 1 (a1)	0.39 +/- 0.49
Airmass coefficient 2 (a2)	1.24 +/- 0.41
Scatter in the residuals of the lightcurve fit is	201.54 %
Best Comparison Star	#1 - [589, 162]
Optimal Aperture	0.75
Optimal Annulus	4.0
Transit Duration (day)	0.064 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1 ± 0.049 vs 0.131 (deviation 0.27σ)
Duration fit vs published	0.064 d vs 0.1075 d (deviation 0.51σ)
Mid-time O–C	-4.6 min
Depth SNR	0.9
Residual scatter	201.54 %

Light curve

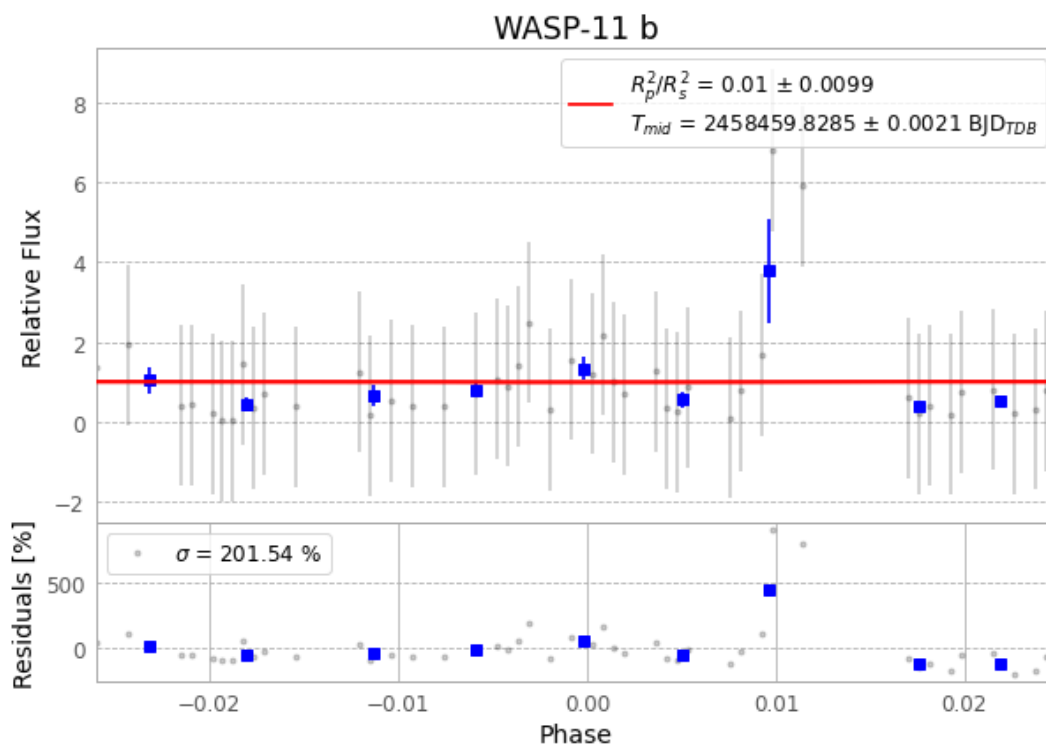


Figure 1 — FinalLightCurve_WASP-11 b_2018-12-06.png (SHA-256 3436c79c64877213...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [587, 370], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad84cc649c084810...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

93 FITS frames (61 MB), HATP-10181207052412.FITS → HATP-10181207100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-181207.log (545888630673...), FinalLightCurve_WASP-11 b_2018-12-06.png (3436c79c6487...), FinalLightCurve_WASP-11 b_2018-12-06.csv (d9da2e7b5dc6...)

Compute resources

Wall time 265.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:52Z.